

Jesse Schmolze

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Education

University of Wisconsin–Madison

B.S. in Economics, Mathematics and Physics

GPA: 3.90/4.00

Relevant Coursework: PhD Econometrics, Stochastic Processes, Brownian Motion and Stochastic Calculus, Data-Driven Stochastic Models and Dynamical Systems, Real Analysis, Introductory Econometrics, Linear Algebra, Machine Learning, Deep Learning, Money and Banking

Awards: Lorraine A. and W. Dwight Stone Scholarship (\$12,000 merit); Bridgewater Forecasting Contest, placed top 2%

Thesis: *Reserve Allocation Under Sanction Risk* (advised by Dr. Louphou Coulibaly)

Madison, Wisconsin

Graduation May 2027

Research Experience

Neural Time-Inhomogeneous Markov Models for Financial Time Series (arXiv:2605.04690)

Madison, Wisconsin

Advised by Prof. Alejandra Quintos, co-authored with Jan Rovirosa

October 2025 – April 2026

- Trained a neural network to model day-to-day equity return shifts, where 99.9% of transitions have under 5 observations; predicted spread moves opposite 21-day realized volatility ($r = -0.62$, $n = 2,347$), so volatile markets narrow next-day outcomes.
- Scaled the model to 10 US bank stocks across 24 years and four crises, where conditioning on volatility instead of returns improved fit 5.6% and beat the baseline in 9 of 10 stocks, with the narrowing effect holding in all 54 windows.
- Presented at the UW–Madison Undergraduate Research Symposium and the ASA Midwest Regional Conference, 2026.

Inspectable Neural Markov Models for Non-Stationary Time Series

Madison, Wisconsin

Co-Authoring Working Paper with Jan Rovirosa, Multi-Asset Extension of Above

January 2026 – June 2026

- Extended the framework to a 10-asset US bank panel spanning 24 years and 6,183 trading days per asset, covering the dot-com bust, the GFC, COVID, and the 2022–23 rate shock, to test how conditioning choice affects coherence across horizons.
- Showed that conditioning on realized volatility rather than returns yields a 5.6% improvement in internal model consistency and superior out-of-sample fit in 9 of 10 assets, identifying the conditioning variable as a first-order modeling decision rather than a notational one.
- Documented a universal cross-asset pattern across 54 model configurations, where high-volatility regimes compress next-step distributions in every configuration tested and the effect is statistically significant in 42 of 54. Coherent dips are visible across all three NBER recessions in the sample.

Who Captures Trade-Network Growth?

Madison, Wisconsin

Research Advised by Prof. Simeon Alder and Prof. Carter Braxton

January 2026 – May 2026

- Estimated how much of partner countries' GDP growth a given country absorbs through its trade network, allowing this absorption rate to vary with the country's institutional quality on a 111-country panel using the Chernozhukov et al. (2018) Double Machine Learning framework.
- Documented a strong institutional moderating gradient where a one-standard-deviation increase in political stability raises the absorption rate by 0.30, with countries in the bottom stability tertile absorbing essentially none of partner growth and countries in the top tertile absorbing 86% of it.
- Stress-tested the result across six governance measures, four estimation methods, polynomial relaxations of the linear specification, and a temporal placebo, confirming the gradient operates at the trade-transmission horizon and is not a spurious institutional-quality channel.

Work & Leadership Experience

Ogborne Capital Management

San Francisco, California

Investment Research Intern

June 2026 – August 2026

- Built an agentic research pipeline in Python (local LLM inference via Ollama, LanceDB retrieval over 2,500 tickers and 10 years of SEC filings) emitting a conviction-ranked shortlist of leads, each requiring a grounded source quote and a falsifiable forward test with explicit confirm/deny conditions.
- Modeled FX risk in the fund's foreign equity book via Monte Carlo simulation of USD/CHF paths under geometric Brownian motion and GARCH(1,1), comparing constant-volatility against volatility-clustering assumptions at the 5th-percentile tail to inform hedging.
- Built a 7-year revenue model for the neocloud computing stack (CRWV, Nebius, IREN, WULF), incorporating contracted backlog, GPU depreciation schedules, and capacity buildout.

Everlight Solar

Brooklyn Center, Minnesota

Business Value Creation Intern

May 2025 – June 2025

- Generated over \$80,000 in revenue across a six-week internship by converting cold residential leads into scheduled appointments through direct in-person outreach.
- Delivered tailored solar value propositions across 500+ household conversations, developing working fluency in the residential solar product set and the financing structures that drive homeowner decisions.
- Iterated on outreach approach across hundreds of homeowner interactions, identifying which opening framings, objection responses, and product angles converted best in different neighborhood demographics, and refining the personal pitch accordingly.

Badger Solar Racing

Madison, Wisconsin

Race Strategy Team Lead

April 2025 – Present

- Built an end-to-end race-strategy stack for the 2026 American Solar Challenge that ingests live telemetry, weather forecasts, and course elevation to produce 5-minute target-speed decisions across an 8-hour race day with the goal of optimizing range relative to energy constraints.
- Trained a neural network approximation of the team's full MATLAB and Simulink vehicle dynamics model. The approximation matches the original simulator at $R^2 \geq 0.95$ on critical state variables and runs roughly 1000× faster, making large-scale strategy search tractable.
- Trained a reinforcement learning agent on the approximation with a custom reward balancing distance gain against energy use, battery health, and control smoothness, then benchmarked the result against a classical constrained-optimization baseline.

Skills & Interests

Programming: Python (NumPy, PyTorch, Pandas), C++, Simulink, MATLAB, Stata

Other Organizations: Federal Reserve Club; Wisconsin AI Safety Initiative; Undergraduate Business Law Association; Putnam Club

Interests: Bullet Chess, 19th Century Russian Literature, Minnesota Timberwolves, Conscious Hip-hop, Collecting Jordans